Lecture 02: Memory Management

Operating Systems

Paging

- Physical memory is split into fixed-size frames.
- Logical memory is split into pages of the same size.
- Page size in this course: 4 KB.

Translation Lookaside Buffer (TLB)

- A small cache of recent page-table entries.
- TLB hit ratio h = hits / total lookups.

Effective Access Time

- EAT = h(t_TLB + t_mem) + (1 - h)(t_TLB + 2 t_mem)
- Example: h = 0.9, t_TLB = 10 ns, t_mem = 100 ns gives EAT = 120 ns.